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DCIT 26 – APPLICATION DEVELOPMENT AND EMERGING TECHNOLOGIES

MODULE 1 REPORT EMERGING TECHNOLOGIES OVERVIEW AND ITS AREAS

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1. Introduction

Technology is continuing to transform the way people study, work, communicate, and produce goods and services. Some technologies evolve gradually while others introduce novel approaches to solving existing problems. These latter technologies are often referred to as emerging technologies. Module 1 discusses the concept of “emerging technology” and challenges the assumption that the term refers only to recent innovations. According to the lecture, emerging technology is not defined by its novelty or time of inception but by its potential to continuously develop, intersect with other technologies, and disrupt existing systems and processes.

This report aims to discuss the concepts presented in Module 1, including the definition and characteristics of emerging technologies, major fields of emerging technology, and the relationship between emerging technology and the Fourth Industrial Revolution. The explanations follow the ideas presented in the lecture and attempt to simplify the definitions and examples provided by the module.

2. Emerging Technologies Defined

In Module 1 begins the discussion of emerging technologies by defining the term as technologies that are ‘emerge’ or ‘are expected to emerge’ in the future and ‘have the potential to fundamentally change the business landscape’. However, as noted by the lecture, the term ‘emerging’ does not necessarily imply that a technology is recent or newly-invented.

An illustrative example provided by the lecture is artificial intelligence, which was first conceptualized in the 1950’s but continues to be physically realized and integrated into diverse fields of human activity. Another way to think about emerging technology is to consider how existing technology can be combined or used in novel ways. With this in mind, it is possible to think of emerging technology not only as recently-developed solutions but as solutions that continue to evolve, intersect with other fields, disrupt existing practices, and whose final form and impact are not yet fully-realized.

3. Characteristics of Emerging Technology

According to Rotolo, Hicks, and Martin (2015), there are five characteristics that can be used to identify emerging technology. As discussed in Module 1, these characteristics are:

3.1. Radical Novelty

Refers to the degree to which technology is innovative or represents a break away from past practices. An emerging technology can take the form of a developing technology (i.e. a technology that is still in the process of being conceptualized or constructed) or involve the use of an existing technology for a different purpose than the one for which it was intended. Moreover, an emerging technology can represent a fundamentally different approach to achieving a similar end from its predecessors. The lecture cites artificial intelligence as an example of an emerging technology with radical novelty because its diverse applications can lead to fundamentally different technological outcomes. In essence, radical novelty indicates

that an emerging technology is not simply an evolutionary step above existing technology but rather represents a **fundamentally different approach to achieving a similar end**.

3.2. Relatively Fast Growth

This characteristic refers to the relative speed at which emerging technology is adopted and integrated into society as well as the extent to which it is present in scientific literature, patents, prototypes, commercial products, and services. The lecture notes that unlike many other technologies that have reached maturity, emerging technology typically experiences a rapid rise in the number of adopters and users. For example, the lecture cites artificial intelligence as an emerging technology whose rate of adoption and overall presence in the marketplace is steadily increasing.

3.3. Coherence

This characteristic refers to the extent to which disparate research areas and technologies are unified and interconnected. Essentially, for emerging technology, the areas of technology that it intersects with are no longer viewed as separate entities but rather as part of a converging and connected field. The lecture gives the example of cloud computing, artificial intelligence, data analytics, and robotics as a connected set of technologies that can be combined to achieve specific ends. Moreover, as suggested by the lecture, the ability to unify disparate research areas is particularly **pertinent to the application of emerging technology in the business sphere**.

3.4. Prominent Impact

Refers to the degree to which an emerging technology impacts society and the economy. Emerging technology typically has a transformative effect on society – it can disrupt entire industries and fundamentally change the way business is conducted. The extent of this impact varies depending on the domain of an emerging technology but, as noted by the lecture, its impact can extend to multiple industries and transform business practices, institutions, human interactions, and modes of knowledge production. Again, artificial intelligence is used as an illustrative example – the lecture suggests that artificial intelligence can impact business, research, medicine, and individual consumers, fundamentally changing the way these areas operate.

3.5. Uncertainty and Ambiguity

As mentioned by the lecture, emerging technology is often associated with uncertainty and ambiguity. The lecture cites the example of artificial intelligence to note that while the potential of artificial intelligence is clear, its actual impact depends on which of its potential applications are further developed and realized. At the same time, different actors and stakeholders can define emerging technology differently – in other words, while some may define the potential of a particular technology, others may dismiss its impact.

4. Major Emerging Technology Areas

As mentioned earlier, emerging technology can take many forms. According to the lecture, some of the most promising emerging technology areas include Augmented Reality (AR), Artificial Intelligence (AI), Machine Learning, and Quantum Computers.

4.1. Augmented Reality

Augmented Reality (AR) is a technology that allows for the enhancement of real-world sensory events – for instance, by overlaying visual and sound effects on top of real-world imagery. AR differs from Virtual Reality (VR) in that it does not seek to recreate or simulate reality but rather to enhance it. According to the lecture, AR can be used in various fields – for instance, in-car displays can provide guiding imagery that assists the driver in maneuvering. AR can also be used in retail settings to allow for the previewing of products in one's home or to assist with the organization and retrieval of items in the warehouse. The lecture notes that AR is one of the key technologies of the Fourth Industrial Revolution because it can allow businesses to provide contextually relevant information to consumers and facilitate their decision-making.

4.2. Artificial Intelligence

Artificial Intelligence (AI) is a term that refers to the simulation of human intellectual processes by machines. The lecture notes that AI is a crucial area of emerging technology because it underlines many of the characteristics of emerging technology, including: growth, prominent impact, and uncertainty. AI can be applied in diverse fields of human activity – according to the lecture, the applications of AI range from manufacturing and transportation to education and entertainment. Moreover, as mentioned by the lecture, AI can take diverse physical forms – from humanoid robots to cloud-based applications.

4.3. Machine Learning

Machine Learning is a type of AI that allows for training a machine to perform certain tasks, with the ability to learn from experience being similar to human learning. The lecture discusses a number of applications of machine learning, including: medicine, finance, marketing services, and transportation. For instance, in medicine, machine learning algorithms can be used to facilitate the drug discovery process.

4.4. Quantum Computers

Quantum computers utilize the principles of quantum mechanics to perform calculations. According to the lecture, they can perform certain calculations much faster than traditional computers. The basic unit of a quantum computer is a qubit – a quantum analog of a bit. Unlike bits, however, qubits can occupy more than one state at a given time. The lecture notes that IBM, Google, Rigetti Computing, and IonQ have already built quantum computers in their laboratories, while academia and industry are working to develop quantum software. According to the lecture, one of the potential applications of quantum computers is the simulation of natural processes, including the behavior of matter at the atomic level. Similarly, the lecture suggests that the unique characteristics of quantum mechanics can allow for further materials design and discovery.

5. The Fourth Industrial Revolution

The Industrial Revolution is a term that refers to the major changes in production that occurred as a result of the introduction of new technologies. According to the lecture, the Industrial Revolution is a “systematic and fundamental transformation of manufacturing, with the objective of improving quality and reducing costs through physical and digital integration of all functions.” In other words, it refers to a fundamental change in industrial production that makes it more capital- and technology-intensive, changes the working conditions of labor, and facilitates broader social change. The lecture distinguishes between four stages of the Industrial Revolution – Industry 1.0, 2.0, 3.0, and 4.0.

5.1. Industry 1.0

The First Industrial Revolution occurred in the eighteenth century and introduced mechanical production to replace human labor. The means of production included hydroelectric and steam power.

5.2. Industry 2.0

The Second Industrial Revolution began in the nineteenth century and is characterized by the introduction of electricity as the major source of power. Electricity facilitated large-scale and continuous production and allowed for the creation of more complex and sophisticated machinery.

5.3. Industry 3.0

The Third Industrial Revolution began with the appearance of the first computers. It is characterized by the introduction of electronics and information technology into the production process, which automated many aspects of it. According to the lecture, the Fourth Industrial Revolution began with the advent of the Internet and the introduction of alternate energy sources.

5.4. Industry 4.0

The Fourth Industrial Revolution is taking place at the present time and is characterized by disruptive technologies such as the Internet of Things, robotics, Virtual and Augmented Reality, and Artificial Intelligence. According to the lecture, the Fourth Industrial Revolution transforms the relationships between suppliers, producers, consumers, humans, and machines. In essence, it redefines the production process, replacing many of its elements with digital ones.

6. Building Blocks of Industry 4.0

The lecture discusses several key emerging technologies that can be considered the building blocks of Industry 4.0. They include Big Data Analytics, the Internet of Things (IoT), and Cloud Computing.

6.1. Big Data Analytics

Big Data Analytics is the process of collecting, organizing, and analyzing large sets of data in order to extract valuable information from them. In production and in the organizational context, Big Data Analytics can help facilitate faster and more informed decision-making. The lecture notes that the value of Big Data Analytics lies in its ability to analyze valuable information and

provide answers in real-time. As such, it can be used to support managerial and operational decision-making. In essence, Big Data Analytics facilitates extracting valuable information from data.

6.2. Internet of Things

Internet of Things (IoT) is the concept of connecting objects and systems to the Internet, thereby enabling them to share information and perform various functions. Essentially, it allows for interconnectivity and smart systems. Sensors and wireless technology play a major role in IoT. According to the lecture, IoT facilitates field control – in other words, field devices can communicate with each other as well as with centralized controllers. Moreover, IoT allows for decentralized analytics and decision-making, which means that decisions can be made in real-time and with less lag. In this respect, IoT is crucial for smart systems and interconnectivity.

6.3. Cloud Computing

Cloud computing is the designation for hosting services on the Internet. According to the lecture, cloud technologies can be used to facilitate the performance of production systems and processes. In other words, it allows for data-driven operations. The lecture notes that machine data and functionality can be moved to the cloud in order to make them accessible via the Internet. In essence, cloud computing enables data-driven operations.

6.4. Augmented Reality in Building Blocks of Industry 4.0

AR can be used as a workplace assistant to organize work and facilitate operations. The lecture gives the example of using AR to guide warehouse workers to the products they are looking for or to provide repair instructions for the workers to follow when servicing equipment. According to the lecture, such applications of AR can be considered as workplace assistants that provide contextually relevant information to workers. As such, AR can be used not only for entertainment purposes but for industrial applications as well. Moreover, the lecture notes that businesses will soon be able to use AR to provide workers with real-time information that can assist them in their operations.

7. Total Analysis

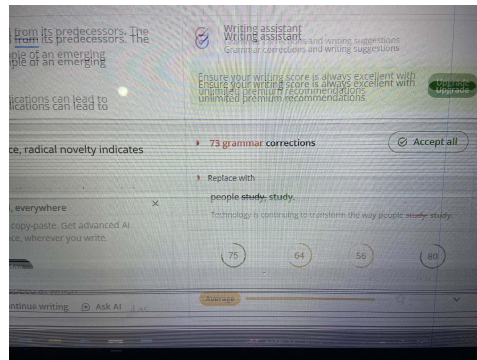
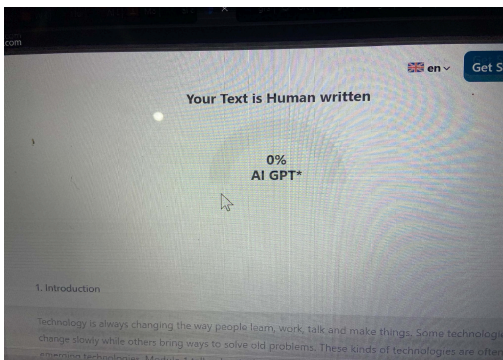
In general, the concepts discussed in Module 1 demonstrate that emerging technologies are connected not only by their novelty but by the fact that they are transformative, impactful, interdisciplinary, and uncertain in their development and outcomes. The examples of AI, AR, machine learning, quantum computing, Big Data Analytics, IoT, and Cloud Computing all suggest that emerging technologies can intersect with one another and facilitate one another's development and application. For instance, IoT produces data, which can then be analyzed using Big Data Analytics. Cloud Computing can also facilitate operational performance by enabling data-driven operations. AI and machine learning, in turn, can be used in diverse fields, including manufacturing, healthcare, finance, and transportation. These examples all suggest that the concepts discussed in Module 1 are closely interconnected. In particular, the lecture suggests that the convergence of these emerging technologies is what makes up the concept of the Fourth Industrial Revolution.

For an IT student, such concepts are valuable because they allow one to develop a more comprehensive understanding of the development and application of technology. More importantly, they suggest that technology should not be considered in isolation but rather as a system that transforms human practices and organizational operations for the better. In essence, Module 1 facilitates both theoretical and practical understanding of the role of technology in human life.

8. Conclusion

In conclusion, emerging technologies are those that are transformative, impactful, interdisciplinary, and uncertain in their development and outcomes. Module 1 highlights the fact that these technologies are not necessarily recent innovations but can also represent disruptive changes to the existing technologies. For instance, AI is considered an emerging technology even though it was first conceptualized in the 1950's – it continues to develop and to be integrated into diverse fields of human activity. The major emerging technologies discussed by the lecture include AR, AI, machine learning, quantum computing, Big Data Analytics, IoT, and Cloud Computing. These technologies are considered building blocks of the Fourth Industrial Revolution because they facilitate technological change and disrupt existing practices. In general, the concepts discussed in Module 1 are integral to understanding the development and application of technology in the modern world.

9. AI Checker and Grammarly Result



10. Reference

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